Photochemical Energy Capture and Overload Stabilization: Functional Bioenergetic Materials After the Core Gate

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Abstract

Papers 1–10 are sufficient to describe a geochemical and protocellular path toward primitive life without requiring modern photosynthesis. This manuscript adds a later expansion channel: photochemical energy capture. The corrected claim is functional rather than molecule-specific. "Chlorophyll-like" means energy-input amplification, not literal chlorophyll at the earliest origin; "melanin-like" means surplus stabilization, not a required first pigment. CDFD predicts that once usable input rises, systems must couple capture to electron transport, proton/ion transport, retention, closure, information stability, and overload buffering. The paper therefore preserves the plant and pigment insights as mature exemplars while keeping the origin-of-life mechanism chemically open.

Keywords: photochemistry; energy capture; overload buffering; prebiotic pigments; origins of life.

Preprint note

This is a computational research article. The companion script produces an illustrative diagnostic reported below. It is not a narrative review and does not claim historical proof of a unique abiogenesis pathway.

Symbol Conventions

See Paper 1, §Symbol Conventions, for the base CDFD and Greek-symbol tables. This paper introduces I_E (usable environmental energy input), σ_e/σ_p (electron/proton conductance proxies), and B_{stab} (net stabilization burden) — the vocabulary Paper 12 assembles into the full tri-regime model.

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1 Why This Paper Exists

Papers 1–10 build the core chain: thermodynamic framing, redox/proton architecture, oscillating polymerization, closure, adaptive boundaries, replication/protocell integration, and parasitic boundary pressure. That sequence can describe chemolithotrophic or geochemical routes toward proto-life [8, 5, 4].

The missing question is different: what happens when a system gains access to a stronger or more spatially independent energy source, especially light? Photochemical input can raise the available driving flux without requiring permanent attachment to one vent geometry. However, naming modern chlorophyll too early makes the theory historically brittle. The corrected principle is:

CDFD predicts functional roles, not mandatory molecules.

2 Functional Material Classes

CDFD role	Symbol	Modern example	Prebiotic candidate class
Energy-input amplification	I_E	Chlorophyll antenna	Fe–S photoredox, porphyrinoids, flavins, quinones, mineral semiconductors
Electron transport	σ_e	ETC cofactors, cytochromes	Fe–S clusters, magnetite-like phases, mixed-valence minerals
Proton/ion coupling	σ_p	Proton motive force	Interfacial water, pores, pH gradients, mineral-water boundaries
Retention	Ξ_{loc}	Membranes, organelles	Mineral pores, gels, coacervates, LLPS droplets
Overload stabilization	B_{stab}	Carotenoids, NPQ, eumelanin/melanins	Polyphenols, disordered aromatics, radical sinks, mineral screens

The names are mnemonic labels for functional channels. They should not be read as claims that those exact substances were present at the earliest transition.

3 Capture Before Stabilization

Let I_E denote usable environmental energy input. A minimal throughput expression is

$$J_{\mathrm{bio}} = I_E \frac{\sigma_e \sigma_p}{B_{\mathrm{stab}}}. \tag{1}$$

Here B_{stab} is a net stabilization burden: radical load, thermal disorder, photodamage risk, hydrolysis pressure, and uncoupled side reactions. A stabilizing material lowers effective B_{stab} only if it absorbs or redirects excess flux without blocking productive electron/proton coupling.

The corrected causal order is therefore

$$\text{capture} \rightarrow \text{couple} \rightarrow \text{retain} \rightarrow \text{close} \rightarrow \text{stabilize surplus}. \tag{2}$$

Chlorophyll-like functions belong at the capture stage. Melanin-like functions belong at the surplus-stabilization stage.

4 Pre-Chlorophyll Photochemical Amplification

Modern chlorophyll is a mature endpoint, not the primitive assumption. Earlier energy-input materials could have performed weaker versions of the same role:

- Fe–S and metal sulfide surfaces that couple light or redox energy to surface reactions,
- semiconducting minerals that create electron-hole separation under illumination,
- porphyrin-like or metal-organic precursors with conjugated absorption,
- flavins, quinones, and aromatic cofactors that shuttle electrons,
- retinal-like or other simple chromophore systems in later protocellular settings.

The historical question is empirical: did any candidate raise I_E while remaining coupled to σ_e, σ_p, and retention?

For this paper, I_E is not total incident light. It is the portion of absorbed or redox input that becomes chemically usable. σ_e measures effective electron-transfer capacity, σ_p measures proton or ion coupling, and B_{stab} measures the burden imposed by photodamage, radical leakage, heating, bleaching, and uncoupled side reactions.

5 Overload Stabilization

Once photon capture or chemical input becomes strong, excess flux becomes dangerous. Productive throughput is limited by the slowest coupled channel:

$$J_{\text{max}} \leq \min(I_E, \sigma_e, \sigma_p, 1/B_{\text{stab}}). \tag{3}$$

If I_E exceeds the capacity of electron transport, proton/ion coupling, carbon fixation, repair, or dissipation, the surplus becomes damage:

$$\Psi_{s,\text{overload}} = \frac{I_E}{C_{\text{use}} + C_{\text{diss}}}. \tag{4}$$

Modern plants solve this with carotenoids, xanthophyll cycling, non-photochemical quenching, repair of PSII, and related dissipative controls. Melanin-like polymers and other disordered aromatics are mature examples of broadband buffering and redox smoothing, not required first molecules [2, 3].

Eumelanin as a modern endpoint. Eumelanin is the strongest modern melanin exemplar for this release because it combines broadband absorption, structural heterogeneity, redox activity, radical buffering, and hydration-dependent electronic behavior [6, 7]. The origin claim remains weaker: prebiotic chemistry only needs a material class that can dissipate surplus input and smooth redox or radical stress without blocking useful electron/proton coupling. Eumelanin is therefore a modern endpoint of the melanin-like function, not a required primordial molecule.

6 Photochemical Fe–S Bridge

Photochemistry need not be late or plant-only. UV-driven prebiotic synthesis of Fe–S clusters suggests that light can participate in forming redox cofactors or cofactor-like clusters before modern photosystems [1]. The corrected order is therefore not

$$\text{dark geochemistry} \rightarrow \text{chlorophyll} \rightarrow \text{plants.}$$

It is

$$\text{redox geochemistry} \leftrightarrow \text{photoredox chemistry} \rightarrow \text{mature photosynthetic pigments.}$$

7 Illustrative Diagnostic

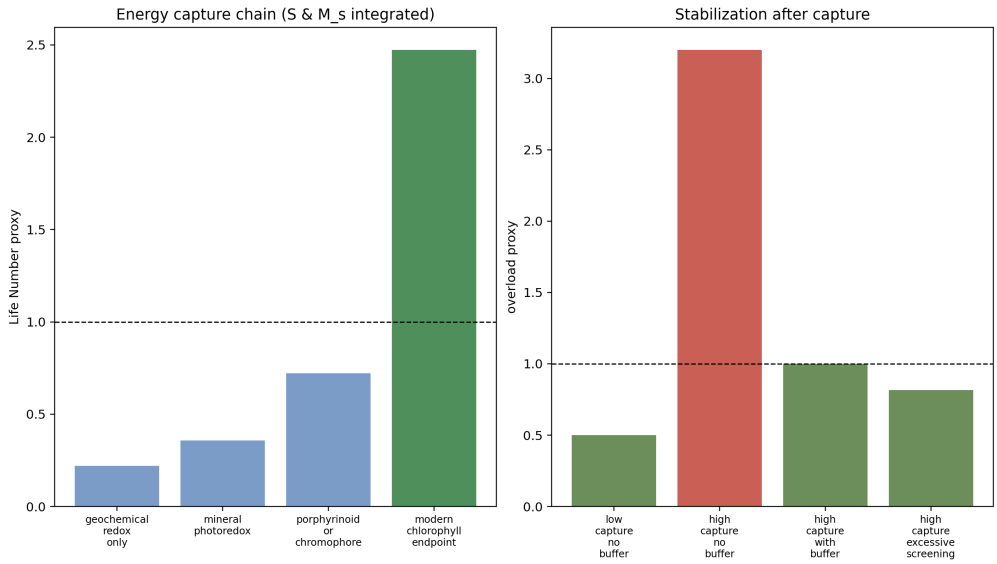

Figure 1: Photochemical capture before surplus stabilization. The first panel treats chlorophyll as the mature endpoint of a wider energy-capture function. The second panel shows why buffering becomes relevant only after captured input creates overload.

8 Falsifiability

The corrected model separates functional claims from material claims:

Layer	Claim	Falsification route
Functional	Proto-life requires sustained input above maintenance	Demonstrate self-maintaining heredity without persistent throughput
Coupling	Electron and proton/ion channels must be co-localized	Demonstrate robust proto-metabolism with uncoupled charge channels
Photochemical	Light-harvesting materials can raise usable throughput	Show photochemical input cannot raise usable redox throughput in plausible settings
Material-specific	Chlorophyll/melanin are examples, not necessities	Replace them with other materials preserving the same function

9 Validation Design

Photochemical validation should separate capture from coupling and coupling from damage control. A minimal experiment would compare illuminated and dark reactors with the same feedstock, pH, ionic strength, compartment type, and mineral or organic surface area. It should then vary one channel at a time: absorber identity for I_E, mixed-valence or Fe–S availability for σ_e, pore or pH-gradient geometry for σ_p, and radical-buffering material for B_{stab}.

The decisive readouts are not color change or total absorption. They are product distribution, redox state, charge separation lifetime, proton/ion-gradient persistence, damage products, and recovery after repeated light cycles. CDFD predicts a non-monotonic surface: extra light helps only when charge transport and buffering prevent overload. If illumination improves products in uncoupled systems just as well as in coupled systems, the coupling claim is weakened. If buffering always suppresses chemistry, rather than improving survival under high input, the overload-stabilization claim is weakened.

10 Conclusion

Photochemical expansion broadens the energy-input landscape; it is not the first step required for life. The release argument is a dependency graph of functions. Capture precedes overload stabilization, and modern materials are illustrative implementations rather than required historical molecules.

11 Reproducibility Note

The companion script `supplementary_ool_11.py` writes the figure under `outputs/paper_11/` plus CSV/JSON summaries. No notebook is required; the static figure and small CSV/JSON files are sufficient.

Related literature

This manuscript is self-contained within the Part II origins-of-life sequence. Shared CDFD/CDFL symbols (Φ, C, S, M_s, Ψ_s) are used as bookkeeping only; they do not require Part I physics results. Field anchors are cited in the body and bibliography.

Competing Interests

The author declares no competing interests.

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Data and Code Availability

Companion scripts, generated figures, and numerical outputs for this manuscript are included in the public CDFD Part II release archive (`https://github.com/bampita-bico/CDFD-Part-II-Release`; Zenodo DOI `10.5281/zenodo.20264779`). See the Reproducibility Note above for the paper-local script path.

Author Contributions

Conceptualization, formal analysis, computational diagnostics, and manuscript preparation: Steve Bico Mujjabi

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